To close the system (8.7)-(8.9) one needs an equation of state, which relates the pressure and temperature to the other variables. For many compressible fluids under moderate conditions, the *ideal gas law* holds. For a *termally* perfect gas one may write

$$p = \rho R T, \tag{8.12}$$

where $R = c_p - c_v$ is the specific gas constant, and c_p and c_v are the specific heats at constant pressure and volume, respectively. For a *calorically* perfect gas the specific heats are constant, and there holds

$$e_{int} = c_v T.$$

Combining these relations leads to

$$\begin{aligned} p &= \rho R T, \\ p &= \rho(c_p - c_v)\frac{e_{int}}{c_v}, \\ p &= (\gamma - 1)\left\{E - \frac{1}{2}\rho \mathbf{u}\cdot\mathbf{u}\right\}, \end{aligned} \tag{8.13}$$

where the definition of the total energy has been used. Furthermore, we have introduced the ratio of specific heats $\gamma = c_p/c_v$. (Note that $\gamma > 1$)

8.2 General Considerations

We have seen in the previous section that the d-dimensional Euler equations governing inviscid compressible fluid flow can be written in the form

$$\mathbf{u}_t + \sum_{j=1}^{d} (\mathbf{f}_j(\mathbf{u}))_{x_j} = 0. \tag{8.14}$$

We avoid the issue of boundary conditions for the moment, and assume that the equation is defined in $(x,t) \in \mathbb{R}^d \times (0,\infty)$. The vector $\mathbf{u}(x) = (u_1(x), \ldots, u_m(x))$ is called the vector of conserved variables, or the state vector. (Note that $m = 5$ for the three-dimensional Euler equations.) Usually the range of the state vector is restricted, i.e. $\mathbf{u} : \mathbb{R}^d \times (0,\infty) \to \mathcal{D} \subset \mathbb{R}^m$. For example, in case of the Euler equations, the density ρ, the Energy E, and the pressure p must all be positive. If we define the momentum variables as $\mathbf{m} = (m_1, m_2, m_3)^T$, where $m_i = \rho u_i$ in the notation of the previous section, we may define the set of admissible sates:

$$\mathcal{D} := \{\mathbf{u} = (\rho, m_1, m_2, m_3, E)^T \in \mathbb{R}^5 : \rho > 0,\ E - \frac{|\mathbf{m}|^2}{2\rho} > 0\}.$$

(Note that the last condition comes from the definition of the pressure (8.13). This condition also implies $E > 0$.).

We are now ready to give a general definition of a hyperbolic system of conservation laws:

Definition 8.1 *The system of conservation laws* (8.14) *is called* hyperbolic *if for any real vector* $\alpha = (\alpha_1, \ldots, \alpha_d)^T \neq 0$, *and any admissible state* $\mathbf{u} \in \mathcal{D}$, *the linear combination* $A(\mathbf{u};\alpha) := \sum_{j=1}^{d} \alpha_j A_j(\mathbf{u})$, *where* A_j *is the Jacobian of* $\mathbf{f}_j$, *may be diagonalized, and all eigenvalues are real. If, in addition, all eigenvalues are distinct, the system is called* strictly hyperbolic.

This definition includes as a special case a one-dimensional scalar conservation law (d=m=1). Consider the Cauchy problem

$$u_t + (f(u))_x = 0, \qquad (x,t) \in \mathbb{R} \times (0,\infty), \tag{8.15}$$

$$u(x,0) = u_0(x). \tag{8.16}$$

We will generally assume that f is a smooth function. We call a *classic* solution of (8.15), (8.16) any differentiable function that satisfies the conservation laws at each (x,t), as well as the initial conditions at $t=0$. The main point of this section is to show that such classic solutions are not the only meaningful solutions for this problem. For the moment however, we assume that u is such a solution. We may write (8.15) as

$$u_t + a(u)u_x = 0. \tag{8.17}$$

where $a(u) = f'(u)$. The interpretation of such an equation is similar to that of the advection equation: Define $\nabla u := (u_t, u_x)^T$, and note that (8.17) implies $\nabla u \cdot d(u) = 0$, where $d(u) = (1, a(u))^T$. The gradient vanishes in direction $d(u)$. In contrast to the linear case, however, $d(u)$ now depends on the solution!

Imagine a parametric curve having at each point (x,t), through which the curve passes, the tangential vector $d(u)$. Assume that locally we can write this curve as a function $x(t)$. Then, $a(u(x(t),t))$ is the slope. The curve $x(t)$ is thus characterized by the ordinary differential equation

$$\frac{dx(t)}{dt} = a(u(x(t),t)).$$

For example, it will be possible to write this locally in a neighborhood of the initial conditions as long as $a(u_0) < \infty$.) We shall call the curves $x(t)$ *characteristics*. By chain rule, we have for the rate-of-change along these curves,

$$\begin{aligned}\frac{du(x(t),t)}{dt} &= \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}\frac{dx}{dt} \\ &= \frac{\partial u}{\partial t} + a(u(x(t),t))\frac{\partial u}{\partial x} = 0.\end{aligned}$$

Hence the solution is constant on the characteristics, and consequently $a(u(x(t),t))$ is constant also. Thus we have obtained that the characteristics are straight lines. This is true in particular at $t=0$, where $a(u(x,0)) = a(u_0(x))$. The characteristic line emanating from the point x_0 can be written

$$x(t) = x_0 + a(u_0(x_0))t.$$

This means that the initial conditions are transported along these lines. Recall that the linear advection equation showed a similar behavior. Note, however, that in the nonlinear case the slopes of the characteristics depend on the initial conditions! In particular, while the characteristics will be straight lines, the slopes will be different at different locations. In particular, the characteristics may intersect!

Example 8.2 (Burgers' Equation) Let $f(u) = \frac{1}{2}u^2$, and define the following initial value problem:

$$\begin{aligned} u_t + \frac{1}{2}\left(u^2\right)_x &= 0 && (x,t) \in \mathbb{R} \times (0,\infty) \\ u(x,0) &= u_0(x) && x \in \mathbb{R} \end{aligned} \tag{8.18}$$

This particular conservation law goes by the name of (inviscid) Burgers equation. We let

$$u_0(x) = \frac{\pi}{2} - \operatorname{atan}(x).$$

Using $f'(u) = a(u) = u$ we see tat the slopes of the characteristics emanating from the point $(x_0, 0)$ have the slope $a(u_0(x_0)) = u_0(x_0)$. Figure 8.3 shows a graphical illustration of the initial conditions and the corresponding characteristics. We note that the characteristics intersect at some point!

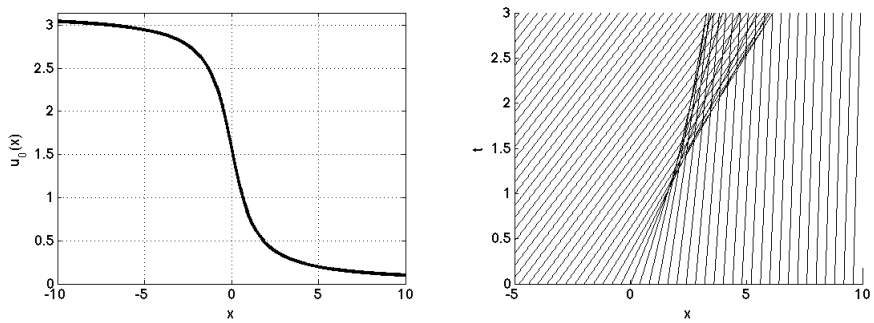


Figure 8.3: Example of characteristics for one-dimensional hyperbolic equations. (Here $f(u) = \frac{1}{2}u^2$ has been chosen, so that $a(u) = u$.) The figure on the left shows a plot of the initial conditions $u_0(x) = \frac{\pi}{2} - \text{atan}(x)$. The figure on the right shows the corresponding characteristics in the $x - t$ plane. The slope of each characteristic line $x(t)$ emanating from a point x_0 has slope $u_0(x_0)$. Thus characteristic lines for smaller values of x have larger slope, and eventually "overtake" those of larger values of x.

What does it mean when the characteristics intersect? It means that we have a serious problem, and our solution breaks down: Since the solution must be constant on each characteristic line, two characteristics meeting at one point provide conflicting information as to what the solution should be. It turns out that we must broaden our notion of what a solution is in order to overcome this problem. In particular, it turns out that we must admit discontinuous solutions.

8.3 Weak solutions

With the definitions of state and flux vector from the previous section, consider the initial value problem

$$\mathbf{u}_t + \sum_{i=1}^{d} \mathbf{f}_i(\mathbf{u})_{x_i} = 0, \qquad x \in \mathbb{R}^d, t > 0 \tag{8.19}$$

$$\mathbf{u}(x,0) = \mathbf{u}_0(x), \qquad x \in \mathbb{R}^d. \tag{8.20}$$

Let $\mathbf{u}$ be a smooth function (at least C^1) satisfying (8.19), (8.20). Multiply (8.14) by a smooth function $\mathbf{v}(x,t)$ having compact support[3] in $\mathbb{R}^d \times [0,\infty)$. Using integration by parts, we can write

$$\begin{aligned} & -\int_0^\infty \int_{\mathbb{R}^d} \left(\mathbf{u}_t + \sum_{i=1}^{d} \mathbf{f}_i(\mathbf{u})_{x_i} \right) \cdot \mathbf{v}\, dxdt = 0 \\ \Leftrightarrow \quad & \int_0^\infty \int_{\mathbb{R}^d} \left(\mathbf{u} \cdot \mathbf{v}_t + \sum_{i=1}^{d} \mathbf{f}_i(\mathbf{u}) \cdot \mathbf{v}_{x_i} \right) dxdt + \int_{\mathbb{R}^d} \mathbf{v}(x,0) \cdot \mathbf{u}_0(x)\, dx = 0 \ . \end{aligned} \tag{8.21}$$

(It is convenient to use functions $\mathbf{v}$ with compact support, so that the boundary terms at $|x| \to \infty$ vanish.) Note that $\mathbf{u}$ does not have to be differentiable for the last line to make sense. This will be the way for us to extend the notion of a solution to a case where $\mathbf{u}$ is discontinuous. Since we have not specified $\mathbf{f}$ beyond assuming it is a smooth function, expression (8.21) makes sense (i.e. the integrals exist) for functions $\mathbf{u}$ that are at least locally bounded[4].

[3] Compact support means that the solution vanishes outside a closed and bounded (i.e. compact) set contained in $\mathbb{R}^d \times [0,\infty)$. We use a subscript "0" to denote compact support for functions that are at least continuous. For example, $C_0^k(\Omega)$ are k times continuously differentiable functions with compact support in domain Ω.

[4] That is, u should be bounded on any compact set contained in the domain of definition. We say $u \in L^\infty_{loc}$. (We can actually allow u to become unbounded for $|x| \to \infty$, because the test functions v have compact support.)

Definition 8.2 *Consider the initial value problem (8.14), and let* $\mathbf{u}_0 \in (L^\infty_{loc}(\mathbb{R}^d)))^m$. *We say* $\mathbf{u} \in (L^\infty_{loc}(\mathbb{R}^d \times [0,\infty))^m$ *is a weak solution to (8.14) if* $\mathbf{u}$ *satisfies (8.21) for any* $\mathbf{v} \in (C^1_0(\mathbb{R}^d \times [0,\infty)))^m$.

Of course, the one-dimensional case is obtained by setting $d = 1$ in (8.21). At this point it is clear that a traditional (strong) solution that satisfies (8.14) pointwise is also a weak solution. (We have just shown this using integration by parts.) However, if one takes (8.21) as a definition of a weak solution, it is likewise clear that a weak solution is *not* necessarily a strong solution, because a function $\mathbf{u}$ need not be differentiable to be a well-defined weak solution. We shall see, however, that at least locally, whenever $\mathbf{u}$ is smooth in a certain region, we get back the strong solution from the weak solution. We just need to examine what happens at points where $\mathbf{u}$ is not smooth. The information contained in the weak form of the equations actually determines what should happen at such points. We first note the following definition:

Definition 8.3 *We call a function* piecewise *smooth in the* (x,t) *plane if there exist a countable number of orientable surfaces, i.e. surfaces permitting piecewise orientation using a normal* $n = (n_{x_1}, n_{x_2}, \ldots, n_{x_d}, n_t)$, *outside of which the solution is smooth, and across which the solution has a jump discontinuity.*

Theorem 8.1 (Rankine-Hugoniot Condition) *Let* $\mathbf{u}$ *be piecewise smooth (as defined above). For any point (x,t) on a surface of discontinuity define* $\mathbf{u}_\pm = \lim_{\varepsilon\to 0^+} \mathbf{u}((x,t) \pm \varepsilon n)$, *where* n *is the normal associated with the surface.Then* $\mathbf{u}$ *is a weak solution to the conservation law (8.14) if and only if* $\mathbf{u}$ *is a classical solution in regions where it is differentiable, and furthermore,* $\mathbf{u}$ *satisfies*

$$(\mathbf{u}_- - \mathbf{u}_+)n_t + \sum_{i=1}^{d} (\mathbf{f}_i(\mathbf{u}_-) - \mathbf{f}_i(\mathbf{u}_+))n_{x_i} = 0 \tag{8.22}$$

across jump discontinuities. Eq. (8.22) is known as the Rankine-Hugoniot condition.

Proof Let $\mathbf{u}$ be a weak solution to (8.14). Choose a control volume $\mathcal{B}'$ that excludes points with $t = 0$, as well as any surface of discontinuity. (See the illustration in Fig. 8.4.) Since $\mathbf{u}$ is smooth in $\mathcal{B}'$, we can pick in (8.21) some function $\mathbf{v}$ having compact support in $\mathcal{B}'$, and integrate by parts, to obtain

$$-\int_{\mathcal{B}'} \left(\mathbf{u}_t + \sum_{i=1}^{d} \mathbf{f}_i(\mathbf{u})_{x_i} \right) \cdot \mathbf{v}\, dxdt = 0.$$

This yields immediately that if $\mathbf{u}$ is a weak solution in $\mathcal{B}'$, it must be a strong solution, because the integral must vanish for integration over any ball that does not include a discontinuity, and for any smooth $\mathbf{v}$ with compact support inside that ball. This is possible only if the term in brackets vanishes.

Now consider a surface of discontinuity $\mathcal{S}$. Choose a control volume $\mathcal{B}$ that excludes $t = 0$, but *includes* the discontinuity, such that the only discontinuity in $\mathcal{B}$ is given by $\mathcal{S} \cap \mathcal{B}$. This divides $\mathcal{B}$ into two subdomains $\mathcal{B}_\pm$, such that $u_\pm \in \mathcal{B}_\pm$. (See again Fig. 8.4.) Since $\mathbf{u}$ is a weak solution, it satisfies (8.21) for the control volume $\mathcal{B}$ with appropriately chosen smooth test

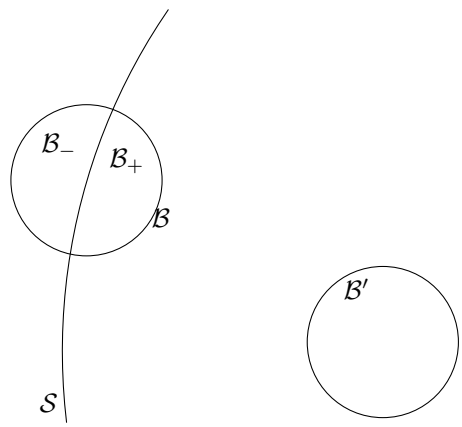


Figure 8.4: Illustration of integration regions used in the proof of Theorem 8.1

functions with compact support on $\mathcal{B}$. Integrating (8.21) by parts in regions $\mathcal{B}_-$ and $\mathcal{B}_+$, where the solution is smooth, one obtains

$$-\int_{\mathcal{B}_-}\left(\mathbf{u}_t+\sum_{i=1}^d \mathbf{f}_i(\mathbf{u})_{x_i}\right)\cdot\mathbf{v}\,dxdt+\int_{\mathcal{S}\cap\mathcal{B}}\mathbf{v}\cdot(\mathbf{u}_-n_t+\sum_{i=1}^d \mathbf{f}_i(\mathbf{u}_-)n_{x_i})\,dA$$
$$-\int_{\mathcal{B}_+}\left(\mathbf{u}_t+\sum_{i=1}^d \mathbf{f}_i(\mathbf{u})_{x_i}\right)\cdot\mathbf{v}\,dxdt-\int_{\mathcal{S}\cap\mathcal{B}}\mathbf{v}\cdot(\mathbf{u}_+n_t+\sum_{i=1}^d \mathbf{f}_i(\mathbf{u}_+)n_{x_i})\,dA=0\ .$$

Note that the boundary integrals over $\mathcal{B}\cap\mathcal{S}$ are the only non-zero boundary integrals, because $\mathbf{v}$ has compact support in $\mathcal{B}$. The minus sign in the second surface integral comes from the fact that the normal used there must be "outward pointing", i.e. from $\mathcal{B}_+$ to $\mathcal{B}_-$, which is exactly opposite to the normal $(n_{x_1},\ldots,n_t)$ that points from $\mathcal{B}_-$ to $\mathcal{B}_+$. The volume integrals vanish, because the weak solution is also a strong solution in smooth regions, while the remaining terms can be rearranged to give (8.22). This uses again the argument that the control volume $\mathcal{B}$ is arbitrary, and the integral must vanish for any $\mathbf{v}$ with compact support in $\mathcal{B}$.

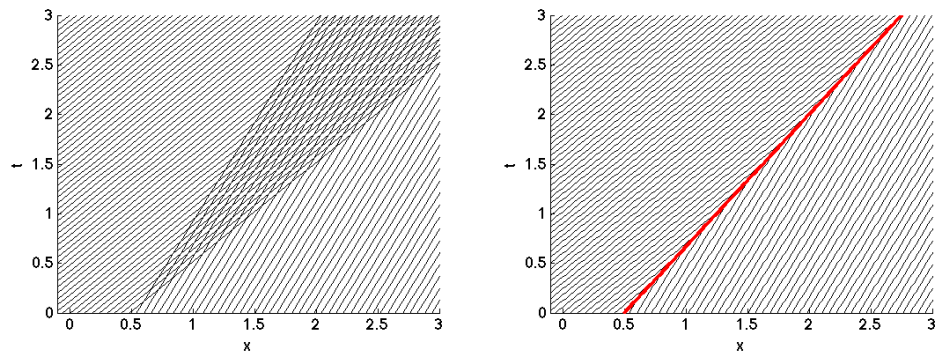
By reversing the calculation one sees immediately that conversely any piecewise smooth function that satisfies (8.14) in smooth regions, and (8.22) across discontinuities is a weak solution. ■

We look at scalar (m=1), one-dimensional (d=1) examples to demonstrate how one may, in simple enough cases, construct weak solutions from the theory developed thus far. First we note that the weak solution is a strong solution whenever u is smooth. We can thus construct a solution using the characteristics, as long as they don't intersect.

Example 8.3 Consider again a Cauchy problem for the Burgers Equation, for which $f(u)=\frac{1}{2}u^2$:

$$u_t+\frac{1}{2}(u^2)_x = 0\ , \qquad (x,t)\in\mathbb{R}\times(0,\infty) \tag{8.23}$$

$$u(x,0) = \begin{cases} u^l\ , & x<x_0\ , \\ u^r\ , & x>x_0\ . \end{cases} \tag{8.24}$$



(a) Intersecting characteristics

(b) A "shock" instead of intersecting characteristics

Figure 8.5: The Burgers Equation: Characteristics for the Riemann problem of Example 8.3 with $x_0 = 0.5$.

This type of problem is often called a *Riemann* problem. For the Burgers equation the slopes of the characteristics are given by $a(u) = f'(u) = u$. As we have seen, those slopes are constant for scalar conservation laws, and we may start constructing the solution from the initial conditions where the slope of the characteristic line emanating from a point, say x_0, will be set by $a(u(x_0, 0)) = u(x_0, 0) = u_0(x_0)$.

We assume that $u^l > u^r$. The characteristics intersect for any $t > 0$, as already the initial conditions are not smooth (cf. Figure 8.5a). From the Rankine-Hugoniot condition, noting $u_- = u^l$ and $u_+ = u^r$, we get

$$\begin{aligned} -(u^l - u^r)n_t &= \frac{1}{2}((u^l)^2 - (u^r)^2)n_x \\ \Longleftrightarrow -(u^l - u^r)n_t &= \frac{1}{2}(u^l - u^r)(u^l + u^r)n_x \\ \Longleftrightarrow -\frac{n_t}{n_x} &= \frac{1}{2}(u^l + u^r) \end{aligned}$$

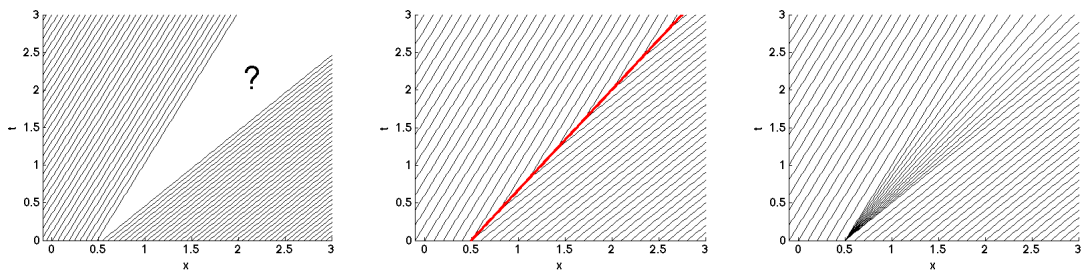
Hence, a discontinuous solution that includes a shock is a weak solution that satisfies the Rankine-Hugoniot condition if

$$u(x,t) = \begin{cases} u^l \ , & x < x_0 + st \ , \\ u^r \ , & x > x_0 + st \ , \end{cases} \tag{8.25}$$

where $s = -\frac{n_t}{n_x} = \frac{1}{2}(u^l + u^r)$ is the shock speed. (Note that $-\frac{n_t}{n_x}$ is nothing but the local slope of the discontinuity in the $x - t$ plane when written as a curve $x(t)$.) This solution is shown in Figure 8.5b.

It may be comforting that we can easily construct a proper weak solution. But the situation is already not so clear if we do the same example for $u^l < u^r$:

Example 8.4 Consider again problem (8.23), (8.24), but this time with $u^l < u^r$. Figure 8.6a shows a plot of the characteristics. We see that the characteristics diverge, and we again have a problem, in that there seems to be a" vacuum" in between. But note that for any $u^l \neq u^r$,



(a) Diverging characteristics (b) A "shock" instead of diverging characteristics (c) A smooth solution

Figure 8.6: The Burgers Equation: Characteristics for the Riemann problem of Example 8.4 with $x_0 = 0.5$.

the discontinuous solution (8.25) with $s = \frac{1}{2}(u^l + u^r)$ is a valid weak solution! Figure 8.6b shows this solution for the case $u^l < u^r$. On the other hand it may be seen that for $u^l < u^r$ a *continuous* solution to (8.23), (8.24) can be constructed (convince yourself that $u = x/t$ satisfies the Burgers equation!):

$$u(x,t) = \begin{cases} u^l \ , & x - x_0 < u^l t \ , \\ (x - x_0)/t \ , & u^l t \leq x - x_0 \leq u^r t \ , \\ u^r \ , & x - x_0 > u^r t \ . \end{cases}$$

Figure 8.6c shows a plot of this solution. So clearly, in this case, we can have more than one solution!

Example 8.4 clearly demonstrates that the class of weak solutions is too large. We have proved by counterexample that weak solutions are not in general unique. (Recall that a well-posed problem needs to have a unique solution.) Both solutions, which are illustrated in fig. 8.6 in terms of the corresponding characteristics are perfectly acceptable weak solutions to the same problem! Obviously then, weak solutions are not unique. We need additional conditions to identify meaningful solutions.